جامعة الإمام عبد الرحمن بن فيصل
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Project Proposal

CIS 517: Data Mining and Data Warehousing (First Semester)

STUDENT NAME	ID

Project Title: Predicting High-Risk Traffic Accident Locations Using Data Mining

1. PROJECT DEFINITION:

This project aims to apply data mining methodologies to identify and predict high-risk traffic accident locations. We won't just analyze historical incident data; we plan to integrate a wider range of influential factors, such as environmental conditions (like weather), road infrastructure details, traffic congestion, and the time of day.

Our goal is to move beyond simply identifying accident hotspots. By combining these diverse data sources, we will build a predictive model that can uncover hidden patterns and relationships contributing to accident frequency and severity. The true value of this project lies in its ability to provide proactive insights to relevant authorities, enabling them to make data-driven decisions for developing effective traffic safety strategies and allocating resources to reduce future accidents.

2. PROJECT OBJECTIVES:

- Establish a structured framework for collecting, preprocessing, and organizing traffic accident data to support analytical processes.
- Employ classification methodologies for assessing the severity of traffic incidents based on influential variables.
- Utilize clustering techniques to detect geographic zones with elevated accident rates and potential risk concentrations.
- Apply frequent patterns to uncover recurring conditions and interdependencies contributing to traffic accidents.
- Construct a predictive model designed to identify locations with a heightened probability of severe incidents.

3. DESCRIBE THE PROBLEM:

The primary challenge addressed in this study is the identification of traffic accident hotspots and the underlying factors that contribute to elevated risk levels. This research employs a combination of data mining techniques to analyze traffic incident data and uncover meaningful patterns.

- **Classification** methods are utilized to evaluate accident severity based on various input features such as location, weather conditions, road type, and time of occurrence.
- **Clustering** techniques are applied to segment geographic areas into accident-prone and low-risk zones, facilitating spatial risk assessment.

- **Frequent Pattern Mining** is used to detect recurring combinations of conditions—such as adverse weather and nighttime highway travel—that are statistically associated with severe accidents.

The integration of these approaches supports the development of a comprehensive risk map, which can serve as a strategic tool for traffic authorities and policymakers in designing targeted safety interventions and resource allocation plans.

4. SIGNIFICANCE AND VALUE:

The analysis of traffic accident data presents a complex problem due to its high dimensionality, heterogeneity, and the presence of numerous interrelated factors. Traditional statistical approaches often fail to capture the hidden structures within such datasets. Data mining techniques provide a systematic methodology for extracting meaningful knowledge, enabling researchers to move beyond descriptive statistics toward predictive and prescriptive insights.
The significance of this project lies in its ability to:

- Demonstrate the application of **classification models** for predicting accident severity based on historical attributes.
- Employ **clustering methods** to reveal natural groupings of accident-prone areas, offering a data-driven approach to hotspot identification.
- Utilize **frequent pattern mining** to uncover recurring factor combinations (e.g., temporal, environmental, and locational variables) that contribute to accident risk.

By integrating these methods, the project advances the understanding of how data mining can be applied to large-scale accident records, thereby providing a framework for knowledge discovery in complex and dynamic domains. The value of this research lies not in operational enforcement but in contributing to the academic field of data mining through the demonstration of practical techniques, comparative evaluation of algorithms, and the generation of insights that may serve as a foundation for future analytical studies.